

KRISTIANSTAD OSTERLEN, Sweden

WMO: 026510

Lat: 55.922N

Lon: 14.086E

Elev: 23

StdP: 101.05

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-8.8	-13.2	1.2	-9.5	-10.9	1.5	-7.5	12.2	6.3	11.2	5.4	1.8	290	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	26.9	18.8	25.1	18.1	23.8	17.7	20.4	24.4	19.4	23.2	18.5	22.1	4.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	21.9	18.0	13.0	21.1	17.1	12.2	20.0	58.3	24.3	55.1	22.9	52.2	22.2	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.8	7.9	-15.4	30.1	5.0	2.2	-19.0	31.7	-21.9	33.0	-24.7	34.2	-28.4	35.8		
(5)			-16.2	21.9	5.1	1.1	-19.8	22.7	-22.8	23.3	-25.7	23.9	-29.3	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	0.1	0.3	2.8	6.8	11.3	14.9	17.6	17.1	13.4	8.9	5.1	2.2	(6)
(7)		DBStd	7.01	4.54	3.80	3.99	3.04	3.11	2.64	2.45	2.28	2.60	3.30	3.55	4.42	(7)
(8)		HDD10.0	1382	306	272	225	102	21	1	0	0	4	61	149	242	(8)
(9)		HDD18.3	3657	564	505	482	346	218	107	43	52	148	294	398	499	(9)
(10)		CDD10.0	805	0	0	1	7	62	148	236	219	107	25	1	0	(10)
(11)		CDD18.3	38	0	0	0	0	0	4	21	13	1	0	0	0	(11)
(12)		CDH23.3	390	0	0	0	1	11	59	216	98	6	0	0	0	(12)
(13)		CDH26.7	57	0	0	0	0	0	9	36	12	0	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.3	4.0	4.1	3.7	3.6	3.7	3.3	3.2	3.5	3.7	3.8	4.1	(14)
(15)	Precipitation	PrecAvg	691	56	46	42	38	50	56	67	64	70	66	73	63	(15)
(16)		PrecMax	867	126	114	85	84	122	128	218	205	171	150	182	152	(16)
(17)		PrecMin	493	1	4	8	10	12	2	3	6	28	17	21	33	(17)
(18)		PrecStd	107	30	22	21	19	25	31	49	45	38	34	34	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	10.1	15.1	20.9	25.0	28.0	30.0	28.2	24.1	17.9	13.2	11.1	(19)
(20)			MCWB	9.0	7.4	9.3	13.0	16.3	18.8	20.1	20.1	18.0	14.8	11.8	10.0	(20)
(21)		2%	DB	8.2	8.2	12.2	17.2	22.2	25.0	27.1	25.9	22.0	16.1	12.0	10.0	(21)
(22)			MCWB	7.4	6.9	7.8	11.0	15.1	17.6	19.3	18.0	16.8	14.0	10.7	9.2	(22)
(23)		5%	DB	7.1	6.9	10.1	15.2	20.2	23.0	25.9	24.0	20.1	14.9	11.0	8.8	(23)
(24)			MCWB	6.4	5.7	7.2	9.8	13.8	16.4	18.6	18.3	15.9	13.2	9.8	8.1	(24)
(25)		10%	DB	6.1	5.8	8.8	13.2	18.2	21.1	24.0	22.8	18.8	13.9	9.9	7.8	(25)
(26)			MCWB	5.3	4.7	6.5	8.8	12.9	15.5	18.0	17.8	15.2	12.4	9.2	7.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	8.5	10.4	13.8	18.1	20.8	21.9	21.4	18.9	15.7	12.5	10.6	(27)
(28)			MCDWB	9.6	9.1	13.4	19.2	22.7	25.8	26.1	25.4	22.9	16.9	13.3	11.2	(28)
(29)		2%	WB	7.5	6.8	8.8	11.9	16.2	18.4	20.6	20.0	17.4	14.5	11.2	9.1	(29)
(30)			MCDWB	8.0	7.9	11.4	16.5	20.4	23.1	25.1	23.3	20.6	15.6	11.9	9.6	(30)
(31)		5%	WB	6.4	5.5	7.5	10.6	14.9	17.3	19.7	19.2	16.5	13.6	10.1	7.9	(31)
(32)			MCDWB	6.9	6.4	9.3	14.0	18.6	21.5	23.8	22.4	19.1	14.7	10.7	8.4	(32)
(33)		10%	WB	5.2	4.4	6.4	9.4	13.7	16.2	18.7	18.3	15.6	12.6	9.1	6.8	(33)
(34)			MCDWB	5.7	5.3	8.1	12.8	17.5	20.0	22.6	21.5	17.9	13.8	9.6	7.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	5.2	7.7	10.8	11.2	10.8	10.5	9.8	9.2	6.9	5.1	4.6	(35)
(36)			MCDBR	4.3	6.3	10.4	15.1	15.0	14.4	14.1	12.3	12.0	8.1	5.5	4.8	(36)
(37)			MCWBR	4.2	4.9	6.9	9.3	8.3	7.7	7.1	6.3	7.5	6.1	4.9	4.6	(37)
(38)		5% WB	MCDWB	4.2	5.6	9.1	13.0	12.5	12.2	11.9	10.5	10.7	7.5	5.2	4.4	(38)
(39)			MCWBR	4.2	4.8	6.5	8.5	7.7	7.2	6.8	6.3	7.2	6.0	4.9	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.334	0.355	0.391	0.388	0.380	0.403	0.404	0.371	0.357	0.337	0.302	(40)	
(41)		taud	2.272	2.298	2.310	2.285	2.343	2.389	2.346	2.349	2.422	2.391	2.327	2.256	(41)	
(42)		Ebn at Noon	592	724	804	826	852	863	833	805	783	692	553	500	(42)	
(43)		Edh at Noon	54	77	97	114	114	111	114	107	87	71	53	44	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.42	1.03	2.44	4.16	5.43	5.78	5.31	4.28	3.00	1.44	0.55	0.29	(44)	
(45)		RadStd	0.08	0.18	0.25	0.49	0.53	0.41	0.62	0.34	0.32	0.23	0.10	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-156	N/A	N/A

Nomenclature: See separate page